9

Thermal Hydraulics

Thermal hydraulics takes into account the temperature variation of a liquid as it flows through the pipeline. This is in contrast to isothermal hydraulics, where there is no significant temperature variation in the liquid. In previous chapters we have concentrated on water pipelines, refined petroleum products (gasoline, diesel, etc.) and other light crude oil pipelines where the liquid temperature was close to ambient temperature. In many cases where heavy crude oil and other liquids of high viscosity have to be pumped, the liquid is heated to some temperature (such as 150°F to 180°F) prior to being pumped through the pipeline. In this chapter we explore how calculations are performed in thermal hydraulics.

9.1 Temperature-Dependent Flow

In the preceding chapters we concentrated on steady-state liquid flow in pipelines without paying much attention to temperature variations along the pipeline. We assumed the liquid entered the pipeline inlet at some temperature such as 70° F. The liquid properties such as specific gravity and viscosity at the inlet temperature were used to calculate the Reynolds number and friction factor and finally the pressure drop due to friction. Similarly, we also used the specific gravity at inlet temperature to calculate the elevation head based on the pipeline topography. In all cases the liquid properties were considered at some constant flowing temperature. These calculations are therefore based on isothermal (constant-temperature) flow.

The above may be valid in most cases in which the liquid transported, such as water, gasoline, diesel, or light crude oil, is at ambient temperature. As the liquid flows through the pipeline, heat may be transferred to or from the liquid from the surrounding soil (buried pipeline) or the ambient air (above-ground pipeline). Significant changes in liquid temperatures due to heat transfer with the surroundings will affect liquid properties such as specific gravity and viscosity. This in turn will affect pressure drop calculations. So far we have ignored this heat transfer effect, assuming minimal temperature variations along the pipeline. However, there are instances when the liquid has to be heated to a much higher temperature than ambient conditions to reduce the viscosity and make it flow easily. Pumping a higher-viscosity liquid that is heated will also require less pump horsepower.

For example, a high-viscosity crude oil (200 cSt to 800 cSt or more at 60°F) may be heated to 160°F before it is pumped into the pipeline. This high-temperature liquid loses heat to the surrounding soil as it flows through the pipeline by conduction of heat from the interior of the pipe to the soil through the pipe wall. The ambient soil temperature may be 40°F to 50°F during the winter and 60°F to 80°F during the summer. Therefore, a considerable temperature difference exists between the hot liquid in the pipe and the surrounding soil.

The temperature difference of about 120°F in winter and 100°F during summer, will cause significant heat transfer between the crude oil and surrounding soil. This will result in a temperature drop of the liquid and variation in liquid specific gravity and viscosity as it flows through the pipeline. Therefore, in such instances we will be wrong in assuming a constant flowing temperature to calculate pressure drop as we do in isothermal flow. Such a heated liquid pipeline may be bare or insulated. In this chapter we will study the effect of temperature variation and friction loss along the pipeline, known as thermal hydraulics.

Consider a 20 in. buried pipeline transporting 8000 bbl/hr of a heavy crude oil that enters the pipeline at an inlet temperature of 160°F. Assume that the liquid temperature has dropped to 124°F at a location 50 miles from the pipeline inlet. Suppose the crude oil properties at 160°F inlet conditions and at 124°F at milepost 50 are as follows:

Temperature (°F)	Specific gravity	Viscosity (cSt)
160	0.9179	40.55
124	0.9306	103.69

Using the 160°F inlet temperature we calculate the frictional pressure drop at inlet conditions to be 7.6 psi/mile. At the temperature conditions at milepost 50, using the given liquid properties we find that the frictional pressure drop has increased to 23.97 psi/mile. Thus, in a 50 mile section of pipe the pressure drop due to friction varies from 7.6 to 23.97 psi/mile as shown in Figure 9.1.

We could use an average value of the pressure drop per mile to calculate the total frictional pressure drop in the first 50 miles of the pipe. However, this will be a very rough estimate. A better approach would be to subdivide the 50 mile section of the pipeline into smaller segments 5 or 10 miles long and to compute the pressure drop per mile for each segment. We will then add the individual pressure drops for each 5 or 10 mile segment to get the total frictional pressure drop in the 50 mile length of pipeline. Of course, this assumes that we have available to us the temperature of the liquid at 5 or 10 mile increments up to milepost 50. The liquid properties and pressure drop due to friction can then be calculated at the boundaries of each 5 or 10 mile segment and the temperature gradient plotted as illustrated in Figure 9.2.

How do we obtain the temperature variation along the pipeline so we can calculate the liquid properties at each temperature and then calculate the pressure drop due to friction? This represents the most complicated aspect of thermal hydraulic analysis. Several approaches have been put forth for calculating the temperature variation in a pipeline transporting heated liquid. We must consider soil temperatures along the pipeline, thermal conductivity of pipe material, pipe insulation (if any), thermal

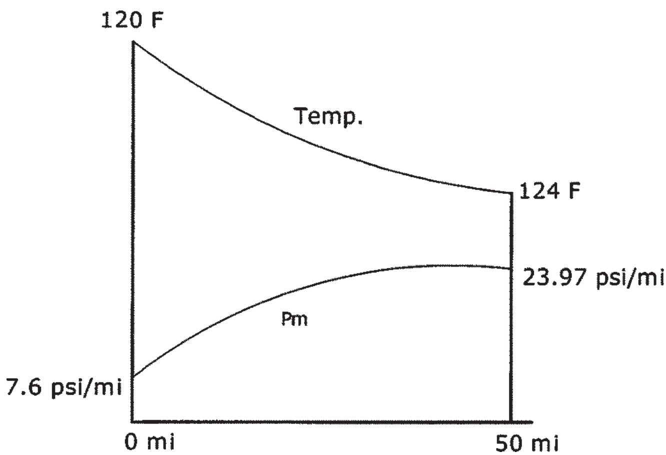

Figure 9.1 Temperature and pressure drop.

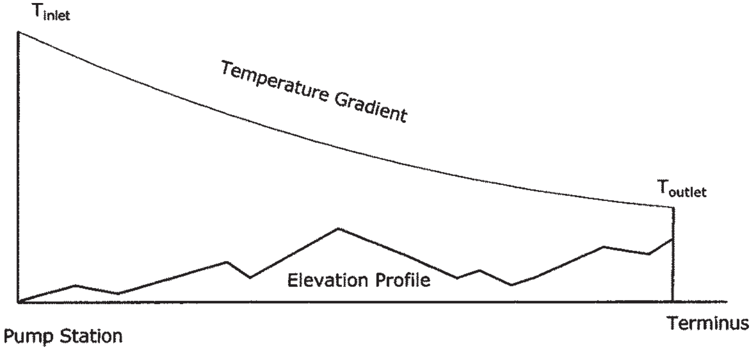

Figure 9.2 Thermal temperature gradient.

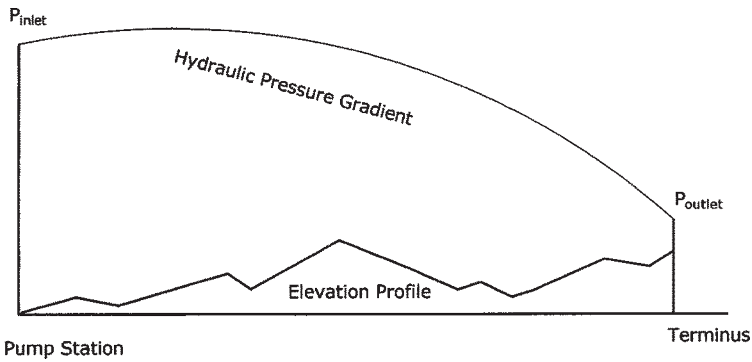

Figure 9.3 Thermal hydraulic pressure gradient.

conductivity of soil, and pipe burial depth. We will present in this chapter a simplified approach to calculating the temperature profile in a buried pipeline. The method and formulas used were developed originally for the Trans-Alaskan Pipeline System (TAPS). These have been found to be quite accurate over the range of temperatures and pressures encountered in heated liquid pipelines today.

The hydraulic gradient showing the pressure profile in a heated liquid pipeline is illustrated in Figure 9.3. Note the curved shape of the gradient, compared with the straight-line gradient in isothermal flow (such as Figures 6.2 or 8.3).

An example will illustrate the use of these formulas. More accurate methods include using a computer software program that will subdivide the pipeline into small segments and compute the heat balance and pressure drop calculations, to develop the pressure and temperature profile for the entire pipeline. One such commercially available program is LIQTHERM developed by SYSTEK (www.systek.us).

9.2 Formulas for Thermal Hydraulics

9.2.1 Thermal Conductivity

Thermal conductivity is the property used in heat conduction through a solid. In English units it is measured in Btu/hr/ft/°F. In SI units thermal conductivity is expressed in W/m/°C.

For heat transfer through a solid of area A and thickness dx, with a temperature difference dT, the formula in English units is:

$$H=K(A)(dT/dx) \tag{9.1}$$

where

H=Heat flux perpendicular to the surface area, Btu/hr
K=Thermal conductivity of solid, Btu/hr/ft/°F
A=Area of heat flux, ft^2
dx=Thickness of solid, ft
dT=Temperature difference across the solid, °F

The term dT/dx represents the temperature gradient in °F/ft. Equation (9.1) is also known as the Fourier heat conduction formula.

It can be seen from Equation (9.1) that the thermal conductivity of a material is numerically equal to the amount of heat transferred across a unit area of the solid material with unit thickness, when the temperature difference between the two faces of the solid is maintained at 1 degree. The thermal conductivity for steel pipe and soil are as follows:

K for steel pipe=29 Btu/hr/ft/°F
K for soil=0.2 to 0.8 Btu/hr/ft/°F

Sometimes heated liquid pipelines are insulated on the outside with an insulating material. The K value for insulation may range from 0.01 to 0.05 Btu/hr/ft/°F.

In SI units, Equation (9.1) becomes

$$H=K(A)(dT/dx) \tag{9.2}$$

where

H=Heat flux, W
K=Thermal conductivity of solid, W/m/°C
A=Area of heat flux, m^2
dx=Thickness of solid, m

dT=Temperature difference across the solid, °C

In SI units, the thermal conductivity for steel pipe, soil and insulation are as follows:

K for steel pipe=50.19 W/m/°C
K for soil=0.35 to 1.4 W/m/°C
K for insulation=0.02 to 0.09 W/m/°C

As an example, consider heat transfer across a flat steel plate of thickness 8 in. and area 100 ft^2 where the temperature difference across the plate thickness is 20°F. From Equation (9.1)

Heat transfer=29×(100)×20×12/8=87,000 Btu/hr

9.2.2 Overall Heat Transfer Coefficient

The overall heat transfer coefficient is also used in heat flux calculations. Equation (9.1) for heat flux can be written in terms of overall heat transfer coefficient as follows:

$$H=U(A)(dT) \tag{9.3}$$

where
U=Overall heat transfer coefficient, Btu/hr/ft^2/°F
Other symbols in Equation (9.3) are the same as in Equation (9.1).
In SI units, Equation (9.3) becomes

$$H=U(A)(dT) \tag{9.4}$$

where
U=Overall heat transfer coefficient, W/m^2/°C
Other symbols in Equation (9.4) are the same as in Equation (9.2). The value of U may range from 0.3 to 0.6 Btu/hr/ft^2/°F in English units and 1.7 to 3.4 W/m^2/°C in SI units.

When analyzing heat transfer between the liquid in a buried pipeline and the outside soil, we consider flow of heat through the pipe wall and pipe insulation (if any) to the soil. If U represents the overall heat transfer coefficient, we can write from Equation (9.3)

$$H=U(A)(T_L-T_S) \tag{9.5}$$

where
A=Area of pipe under consideration, ft^2
T_L=Liquid temperature, °F

T_S=Soil temperature, °F
U=Overall heat transfer coefficient, $Btu/hr/ft^2/°F$

Since we are dealing with temperature variation along the pipeline length, we must consider a small section of pipeline at a time when applying above Equation (9.5) for heat transfer.

For example, consider a 100 ft length of 16 in. pipe carrying a heated liquid at 150°F. If the outside soil temperature is 70°F and the overall heat transfer coefficient

$$U=0.5\ Btu/hr/ft^2/°F$$

then we can calculate the heat transfer using Equation (9.5) as follows:

$$H=0.5\ A\ (150-70)$$

where A is the area through which heat flux occurs:

$$A=\pi\times(16/12)\times 100=419\ ft^2$$

Therefore

$$H=0.5\times 419\times 80=16{,}760\ Btu/hr$$

9.2.3 Heat Balance

The pipeline is subdivided and for each segment the heat content balance is computed as follows:

$$H_{in}-\Delta H+H_w=H_{out} \tag{9.6}$$

where
H_{in}=Heat content entering line segment, Btu/hr
ΔH=Heat transferred from line segment to surrounding medium (soil or air), Btu/hr
H_w=Heat content from frictional work, Btu/hr
H_{out}=Heat content leaving line segment, Btu/hr

In the above we have included the effect of frictional heating in the term H_w. With viscous liquids the effect of friction is to create additional heat which would raise the liquid temperature. Therefore in thermal hydraulic analysis frictional heating is included to improve the calculation accuracy.

In SI units, Equation (9.6) will be the same, with each term expressed in watts instead of Btu/hr.

The heat balance Equation (9.6) forms the basis for computing the outlet temperature of the liquid in a segment, starting with its inlet temperature and taking into account the heat loss (or gain) with the surroundings and frictional heating. In the following sections we will formulate the method of calculating each term in Equation (9.6).

9.2.4 Logarithmic Mean Temperature Difference (LMTD)

In heat transfer calculations, due to varying temperatures it is customary to use a slightly different concept of temperature difference called the logarithmic mean temperature difference (LMTD). The LMTD between the liquid in the pipeline and the surrounding medium is calculated as follows:

Consider a pipeline segment of length Δx with liquid temperatures T_1 at the upstream end and T_2 at the downstream end of the segment. If T_s represents the average soil temperature (or ambient air temperature for an above-ground pipeline) surrounding this pipe segment, the logarithmic mean temperature of the pipe (T_m) segment is calculated as follows:

$$T_m - T_S = \frac{(T_1 - T_S) - (T_2 - T_S)}{\mathrm{Log}_e[(T_1 - T_S)/(T_2 - T_S)]} \tag{9.7}$$

where

T_m=Logarithmic mean temperature of pipe segment, °F
T_1=Temperature of liquid entering pipe segment, °F
T_2=Temperature of liquid leaving pipe segment, °F
T_S=Sink temperature (soil or surrounding medium), °F

In SI units, Equation (9.7) will be the same, with all temperatures expressed in °C instead of °F. For example, if the average soil temperature is 60°F and the temperatures of the pipe segments upstream and downstream are 160°F and 150°F, respectively, the logarithmic mean temperature of the pipe segment is:

$$T_m = 60 + \frac{(160-60)-(150-60)}{\mathrm{Log}_e[(160-60)/(150-60)]} = 60 + 94.88 = 154.88°\mathrm{F}$$

We have thus calculated the logarithmic mean temperature of the pipe segment to be 154.88°F. If we had used a simple arithmetic average we would get the following for the mean temperature of the pipe segment:

Arithmetic mean temperature=(160+150)/2=155°F

This is not too far off the logarithmic mean temperature T_m calculated above. It can be seen that the logarithmic mean temperature approach gives a slightly more accurate representation of the average liquid temperature in the pipe segment. Note that the use of natural logarithm in Equation (9.7) signifies an exponential decay of the liquid temperature in the pipeline segment. In this example, the LMTD for the pipe segment is

LMTD=154.88-60=94.88°F

If we assume an overall heat transfer coefficient U=0.5 Btu/hr/ft^2/°F, we can estimate the heat flux from this pipe segment to the surrounding soil using Equation (9.4) as follows:

Heat flux=0.5×1×94.88=47.44 Btu/hr per ft^2 of pipe area

9.2.5 Heat Entering and Leaving Pipe Segment

The heat content of the liquid entering and leaving a pipe segment is calculated using the mass flow rate of the liquid, its specific heat and the temperatures at the inlet and outlet of the segment. The heat content of the liquid entering the pipe segment is calculated from

$$H_{in}=w(C_{pi})(T_1) \tag{9.8}$$

The heat content of the liquid leaving the pipe segment is calculated from

$$H_{out}=w(C_{po})(T_2) \tag{9.9}$$

where

H_{in}=Heat content of liquid entering pipe segment, Btu/hr
H_{out}=Heat content of liquid leaving pipe segment, Btu/hr
C_{pi}=Specific heat of liquid at inlet, Btu/lb/°F
C_{po}=Specific heat of liquid at outlet, Btu/lb/°F
w=Liquid flow rate, lb/hr
T_1=Temperature of liquid entering pipe segment, °F
T_2=Temperature of liquid leaving pipe segment, °F

The specific heat C_p of most liquids ranges between 0.4 and 0.5 Btu/lb/°F (0.84 and 2.09 kJ/kg/°C) and increases with liquid temperature. For petroleum fluids C_p can be calculated if the specific gravity or API gravity and temperatures are known.

In SI units Equations (9.8) and (9.9) become

$$H_{in}=w(C_{pi})(T_1) \tag{9.10}$$

$$H_{out}=w(C_{po})(T_2) \tag{9.11}$$

where

H_{in}=Heat content of liquid entering pipe segment, J/s (W)
H_{out}=Heat content of liquid leaving pipe segment, J/s (W)

C_{pi}=Specific heat of liquid at inlet, kJ/kg/°C
C_{po}=Specific heat of liquid at outlet, kJ/kg/°C
w=Liquid flow rate, kg/s
T_1=Temperature of liquid entering pipe segment, °C
T_2=Temperature of liquid leaving pipe segment, °C

9.2.6 Heat Transfer: Buried Pipeline

Consider a buried pipeline, with insulation, that transports a heated liquid. If the pipeline is divided into segments of length L we can calculate the heat transfer between the liquid and the surrounding medium using the following equations:

In English units

$$H_b=6.28(L)(T_m-T_s)/(Parm1+Parm2) \tag{9.12}$$
$$Parm1=(1/K_{ins})Log_e(R_i/R_p) \tag{9.13}$$
$$Parm2=(1/K_s)Log_e[2S/D+((2S/D)^2-1)^{1/2}] \tag{9.14}$$

where

H_b=Heat transfer, Btu/hr
T_m=Log mean temperature of pipe segment, °F
T_S=Ambient soil temperature, °F
L=Pipe segment length, ft
R_i=Pipe insulation outer radius, ft
R_p=Pipe wall outer radius, ft
K_{ins}=Thermal conductivity of insulation, Btu/hr/ft/°F
K_s=Thermal conductivity of soil, Btu/hr/ft/°F
S=Depth of cover (pipe burial depth) to pipe centerline, ft
D=Pipe outside diameter, ft

Parm1 and Parm2 are intermediate values that depend on parameters indicated.

In SI units, Equations (9.12), (9.13), and (9.14) become

$$H_b=6.28(L)(T_m-T_s)/(Parm1+Parm2) \tag{9.15}$$
$$Parm1=(1/K_{ins})Log_e(R_i/R_p) \tag{9.16}$$
$$Parm2=(1/K_s)Log_e[2S/D+((2S/D)^2-1)^{1/2}] \tag{9.17}$$

where

H_b=Heat transfer, W
T_m=Log mean temperature of pipe segment, °C
T_s=Ambient soil temperature, °C

L=Pipe segment length, m
R_i=Pipe insulation outer radius, mm
R_p=Pipe wall outer radius, mm
K_{ins}=Thermal conductivity of insulation, W/m/°C
K_s=Thermal conductivity of soil, W/m/°C
S=Depth of cover (pipe burial depth) to pipe centerline, mm
D=Pipe outside diameter, mm

9.2.7 Heat Transfer: Above-Ground Pipeline

An above-ground insulated pipeline can also be used to transport a heated liquid. If the pipeline is divided into segments of length L, we can calculate the heat transfer between the liquid and the ambient air using the following equations.

In English units

$$H_a=6.28(L)(T_m-T_a)/(Parm1+Parm3) \quad (9.18)$$
$$Parm3=1.25/[R_i(4.8+0.008(T_m-T_a))] \quad (9.19)$$
$$Parml=(1/K_{ins})Log_e(R_i/R_p) \quad (9.20)$$

where

H_a=Heat transfer, Btu/hr
T_m=Log mean temperature of pipe segment, °F
T_a=Ambient air temperature, °F
L=Pipe segment length, ft
R_i=Pipe insulation outer radius, ft
R_p=Pipe wall outer radius, ft
K_{ins}=Thermal conductivity of insulation, Btu/hr/ft/°F
In SI units, Equations (9.18), (9.19), and (9.20) become

$$H_a=6.28(L)(T_m-T_a)/(Parm1+Parm3) \quad (9.21)$$
$$Parm3=1.25/[R_i(4.8+0.008(T_m-T_a))] \quad (9.22)$$
$$Parm1=(1/K_{ins})Log_e(R_i/R_p) \quad (9.23)$$

where

H_a=Heat transfer, W
T_m=Log mean temperature of pipe segment, °C
T_a=Ambient air temperature, °C
L=Pipe segment length, m
R_i=Pipe insulation outer radius, mm
R_p=Pipe wall outer radius, mm
K_{ins}=Thermal conductivity of insulation, W/m/°C

9.2.8 Frictional Heating

The frictional pressure drop causes heating of the liquid. The heat gained by the liquid due to friction is calculated using the following equations:

$$H_w=2545\ (HHP) \tag{9.24}$$
$$HHP=(1.7664\times10^{-4})(Q)(Sg)(h_f)(L_m) \tag{9.25}$$

where

H_w=Frictional heat gained, Btu/hr
HHP=Hydraulic horsepower required for pipe friction
Q=Liquid flow rate, bbl/hr
Sg=Liquid specific gravity
h_f=Frictional head loss, ft/mile
L_m=Pipe segment length, miles
In SI units, Equations (9.24) and (9.25) become

$$H_w=1000\ (Power) \tag{9.26}$$
$$Power=(0.00272)(Q)(S_g)(h_f)(L_m) \tag{9.27}$$

where

H_w=Frictional heat gained, W
Power=Power required for pipe friction, kW
Q=Liquid flow rate, m^3/hr
Sg=Liquid specific gravity
h_f=Frictional head loss, m/km
L_m=Pipe segment length, km

9.2.9 Pipe Segment Outlet Temperature

Using the formulas developed in the preceding sections and referring to the heat balance Equation (9.6), we can now calculate the temperature of the liquid at the outlet of the pipe segment as follows: For buried pipe:

$$T_2=(1/wC_p)[2545\ (HHP)-H_b+(wC_p)T_1] \tag{9.28}$$

For above-ground pipe:

$$T_2=(1/wC_p)[2545(HHP)-H_a+(wC_p)T_1] \tag{9.29}$$

where

H_b=Heat transfer for buried pipe, Btu/hr from Equation (9.12)
H_a=Heat transfer for above-ground pipe, Btu/hr from Equation (9.18)
C_p=Average specific heat of liquid in pipe segment

For simplicity, we have used the average specific heat above for the pipe segment based on C_{pi} and C_{po} discussed earlier in Equations (9.8) and (9.9).

In SI units, Equations (9.28) and (9.29) can be expressed as

For buried pipe:

$$T_2=(1/wC_p)\ [1000(\text{Power})-H_b+(wC_p)T_1] \tag{9.30}$$

For above-ground pipe:

$$T_2=(1/wC_p)\ [1000(\text{Power})-H_a+(wC_p)T_1] \tag{9.31}$$

where

H_b=Heat transfer for buried pipe, W
H_a=Heat transfer for above-ground pipe, W
Power=Frictional power defined in Equation (9.27), kW

9.2.10 Liquid Heating due to Pump Inefficiency

Since a centrifugal pump is not 100% efficient, the difference between the hydraulic horsepower and the brake horsepower represents power lost. Most of this power lost is converted to heating the liquid being pumped. The temperature rise of the liquid due to pump inefficiency may be calculated from the following equation:

$$\Delta T=(H/778\ C_P)(1/E-1) \tag{9.32}$$

where

ΔT=Temperature rise, °F
H=Pump head, ft
C_P=Specific heat of liquid, Btu/lb/°F
E=Pump efficiency, as a decimal value less than 1.0

When considering thermal hydraulics, the above temperature rise as the liquid moves through a pump station should be included in the temperature profile calculation. For example, if the liquid temperature has dropped to 120°F at the suction side of a pump station and the temperature rise due to pump inefficiency causes a 3°F rise, the liquid temperature at the pump discharge will be 123°F.

Example Problem 9.1

Calculate the temperature rise of a liquid (specific heat=0.45 Btu/lb/°F) due to pump inefficiency as it flows through a pump. Pump head=2450 ft and pump efficiency=75%.

Solution

From Equation (9.32), the temperature rise is

$$\Delta T=(2450/(778\times0.45))(1/0.75-1)=2.33°F$$

We will now use the equations discussed in this chapter to calculate the thermal hydraulic temperature profile of a crude oil pipeline.

Example Problem 9.2

A 16 in., 0.250 in. wall thickness, 50 mile long buried pipeline transports 4000 bbl/hr of heavy crude oil that enters the pipeline at 160°F. The crude oil has a specific gravity and viscosity as follows:

Temperature (°F):	100	140
Specific gravity:	0.967	0.953
Viscosity (cSt):	2277	348

Assume a pipe burial depth of 36 in. to the top of pipe and 1.5 in. insulation thickness with a thermal conductivity (K value) of 0.02 Btu/hr/ft/°F. Also assume a uniform soil temperature of 60°F with a K value of 0.5 Btu/hr/ft/°F. Using the heat balance equation calculate the outlet temperature of the crude oil at the end of the first mile segment. Assume an average specific heat of 0.45 for the crude oil.

Solution

First calculate the heat transfer for buried pipe using Equations (9.12) through (9.14):

$$Parm1=(1/0.02)Log_e(9.5/8)=8.5925$$
$$Parm2=(1/0.5)Log_e[2\times44/16+((2\times44/16)^2-1)^{1/2}]=4.7791$$
$$H_b=6.28(5280)(T_m-60)/(8.5925+4.7791)$$

or

$$H_b=2479.76(T_m-60)\ \text{Btu/hr} \tag{9.33}$$

The log mean temperature T_m of this 1 mile pipe segment has to be approximated first, since it depends on the inlet temperature, soil temperature, and the unknown liquid temperature at the outlet of the 1 mile segment.

As a first approximation, assume the outlet temperature at the end of the 1 mile segment to be T_2=150. Calculate T_m using Equation (9.7):

$$T_m = \frac{60 + (160 - 60) - (150 - 60)}{\mathrm{Log}_e[(160 - 60)/(150 - 60)]} \tag{9.34}$$

or

$$T_m = 154.91°F$$

Therefore, H_b from Equation (9.33) above becomes

$$H_b = 2479.76(154.91\text{-}60) = 235{,}354 \text{ Btu/hr}$$

The frictional heating component H_w will be calculated using Equations (9.24) and (9.25). The frictional head drop h_f depends on the specific gravity and viscosity at the calculated mean temperature T_m. Using the viscosity-temperature relationship from Chapter 2, we calculate the specific gravity and viscosity at 154.91°F to be 0.9478 and 200.22 cSt, respectively.

$$\text{Reynolds number,} \quad R = \frac{92.24 \times (4000 \times 24)}{15.5 \times 200.22} = 2853$$

Using the Colebrook-White equation, the friction factor is

$$f = 0.034$$

Frictional head drop h_f will be calculated from the Darcy-Weisbach equation (3.26) as follows:

$$h_f = 0.034(5280 \times 12/15.5)(V^2/64.4)$$

The velocity V is calculated using Equation (3.12) as follows

$$V = 0.2859(4000)/(15.5)^2 = 4.76 \text{ ft/s}$$

Therefore, the frictional pressure drop is

$$h_f = 0.034(5280 \times 12/15.5)(4.76 \times 4.76/64.4) = 48.89 \text{ ft}$$

From Equation (9.25) the frictional horsepower (HP) is

$$HHP = (1.7664 \times 10^{-4}) \times 4000 \times 0.9478 \times 48.89 \times 1.0 = 32.74$$

Therefore frictional heating from Equation (9.24) is

$$H_w = 2545 \times 32.74 = 83{,}323 \text{ Btu/hr}$$

The mass flow rate is

$$w=4000\times5.6146\times0.9478\times62.4=1.328\times10^{6}\ \text{lb/hr}$$

From Equation (9.30) the liquid temperature at the outlet of the 1 mile segment is

$$T_2=(1/(1.328\times10^{6}\times0.45))[83{,}323-235{,}354+1.328\times10^{6}\times0.45\times160]$$
$$=[-0.255+160]=159.75°F$$

This value of T_2 is used as a second approximation in Equation (9.34) to calculate a new value of T_m and subsequently the next approximation for T_2. Calculations are repeated until successive values of T_2 are within close agreement. This is left as an exercise for the reader.

It can be seen from the foregoing that manual calculation of temperatures and pressures along a heated oil pipeline is definitely a laborious process, but that can be eased using programmable calculators and personal computers.

Thermal hydraulics is very complex and calculations require utilization of some type of computer program to generate quick results. Such a program can subdivide the pipeline into short segments and calculate the temperatures, liquid properties, and pressure drops as we have seen in the examples in this chapter. Several software packages are commercially available to perform thermal hydraulics. One such package is LIQTHERM developed by SYSTEK Technologies, Inc. (www.systek.us). For a sample output report from a liquid pipeline thermal hydraulics analysis using LIQTHERM software, refer to Table A.15 in Appendix A.

9.3 Summary

We have explored the thermal effects of pipeline hydraulics in this chapter. To transport viscous liquids, they have to be heated to a temperature sometimes much higher than the ambient conditions. This temperature differential between the pumped liquid and the surrounding soil (buried pipeline) or ambient air (above-ground pipeline) causes heat transfer to occur, resulting in temperature variation of the liquid along the pipeline. Unlike isothermal flow, where the liquid temperature is uniform throughout the pipeline, heated pipeline hydraulics requires subdividing the pipeline into short segments and calculating pressure drops based on liquid properties at the average temperature of each segment. We illustrated this using an example that showed how the temperature varies along the pipeline. The concept of LMTD was introduced for determining a more accurate average segment temperature. Also, a method to compute the heat transfer between the liquid and the surrounding medium was shown taking into account thermal conductivities of pipe, soil, and insulation, if present. The heating of liquid due to friction was also quantified, as was the liquid heating associated with pump inefficiency.

It was pointed out that unlike isothermal hydraulics, thermal hydraulics is a complex phenomenon that requires computer methods to correctly solve equations for temperature variation and pressure drop calculations.

9.4 Problems

9.4.1 An 8 in. nominal diameter pipe is used to move heavy crude oil from a heated storage tank to a refinery 12 miles away. The inlet temperature is 180°F and the crude oil has the following characteristics:

Temperature (°F):	100	180
Specific gravity:	0.985	0.912
Viscosity (cSt):	4000	25

If the outlet temperature cannot drop below 150°F, what minimum flow rate needs to be maintained in the pipeline? Use the MIT equation and 0.002 in. pipe roughness, 70°F soil temperature, 36 in. burial depth, and 0.25 in. insulation, with K=0.02 Btu/hr/ft/°F. The K value for soil=0.6 Btu/hr/ft/°F. Pipeline pressures are limited to ANSI 600 rating (1440 psi).

9.4.2 In Problem 9.4.1, if insulation were absent what minimum flow rate could be tolerated? Compare the HP requirements in both cases. Assume 15 psi pump suction pressure.

9.4.3 From Problem 9.4.1 develop a set of data points for flow rate versus pressure required for plotting a system head curve. Assume 50 psi delivery pressure.